

STATS 418

Color Analysis using Machine Learning

Simplifying style and building a cohesive closet

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Overview/Motive

Why color analysis?

Problem: While some enjoy the process, getting dressed is unnecessarily difficult for many from celebrities to everyday people.

Motivation: Your clothes should fit you rather than you needing to fit your clothes. Using color analysis, people can filter which clothes suit them and which do not to optimize the process.



Data Collection Approach



Data Source:

ColorMine.ai – Celebrity color analysis directory



Method:

Web scrape each celebrity's information



Progress:

Collected and cleaned each celebrity's job, job category, eye color, hair color, skin tone, contrast, **color season**, and **best/worst colors**

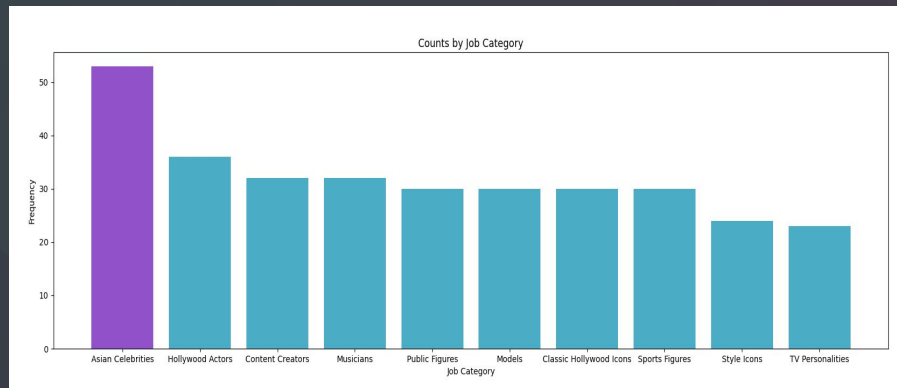
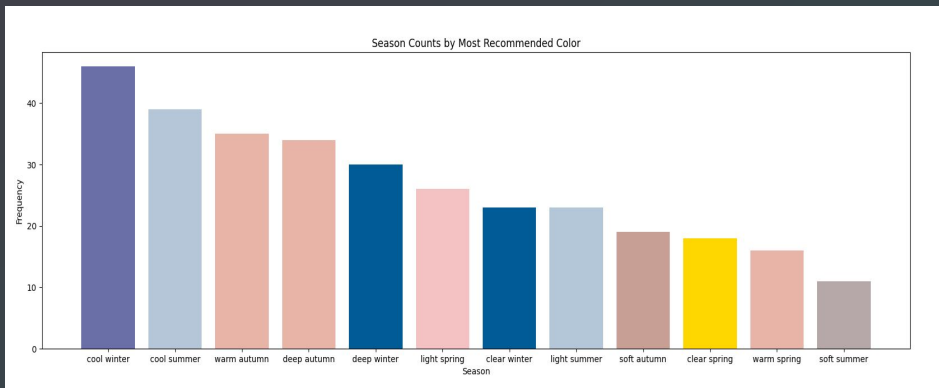
Exploratory Analysis

Key Patterns

- Unbalanced Data
- Overlapping palettes
- Mostly Asian celebrities

Data Processing and Quality

The data was fairly clean with 12 color seasons, but the main cleaning narrows the 137 eye colors to five colors and the 74 hair colors to five as well.



Proposed Model

MLP vs. CNN



Engineered Features:

- Hair color
- Eye color
- Job category



Evaluation Metrics:

- Accuracy
- F1 score
- AUC-ROC
- Loss



Milestones and Timeline:

4/13: choose data	5/2-5/5: EDA
4/20: create environment	5/6-5/17: modeling
4/22-29: get data	5/18-5/26: app
4/30-5/1: clean	5/27-6/2: final presentation

Challenges and Solutions:

- **Small data** → K-fold cv
- **Image inconsistencies** → resize
- **Uneven counts** → change to binary

Remaining Work:

- **Modeling**
- **App**
- **Presentation**

Timeline and Next Steps



Thank You!

Links:

Link to github: https://github.com/jfung07/FungJulia_Stats418_Final

Link to site: <https://colormineai.com/celebrity/>